

1.

$$f(x) = \int_8^{30} 20 \ln \left(\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right) - 9.8t \quad dt$$

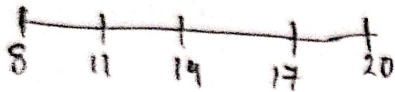
$$a. I = \int_8^{20} f(x) dx$$

$$n = 4$$

$$a = 8$$

$$b = 20$$

$$\therefore h = \frac{20-8}{4} = 3$$



$$f(8) = 177.266$$

$$f(11) = 252.847$$

$$f(14) = 334.244$$

$$f(17) = 422.142$$

$$f(20) = 517.349$$

$$\begin{aligned} \therefore I &= \frac{14-8}{6} [177.266 + 334.244 + 4 \times 177.266] \\ &\quad + \frac{20-14}{6} [334.244 + 517.349 + 4 \times 422.142] \\ &= 3760.735 \end{aligned}$$

$$b. I = \int_{20}^{30} f(x) dx$$

$$a = 20$$

$$b = 30$$

$$n = 2$$

$$h = 5$$

$$f(20) = 517.349$$

$$f(25) = 901.674$$

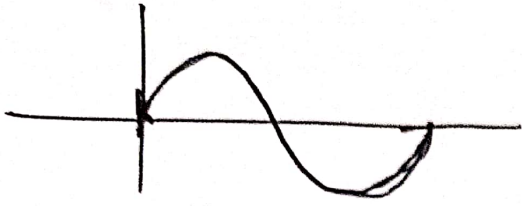
$$f(30) = 695.007$$

$$\begin{aligned} \therefore I &= \frac{25-20}{2} [517.349 + 901.674] + \frac{30-25}{2} [695.007 + 901.674] \\ &= 7021.72 \end{aligned}$$

$$c. I = I_1 + I_2 = 10782.455$$

1805030
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$$f(x) = \sin x.$$



$$\begin{aligned} f(0) &= 0 \\ f(\pi/4) &= 0.707 \\ f(\pi/2) &= 1 \end{aligned} \quad \left\{ \begin{array}{l} x_0 = 0 \\ x_1 = \frac{\pi}{4} \\ x_2 = \pi/2 \end{array} \right.$$

$$\therefore f(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$b_0 = f[x_0] = 0.$$

$$b_1 = \frac{0.707}{\pi/4} = 0.9$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

$$= \frac{0.373 - 0.9}{\frac{\pi}{2}} = -0.335.$$

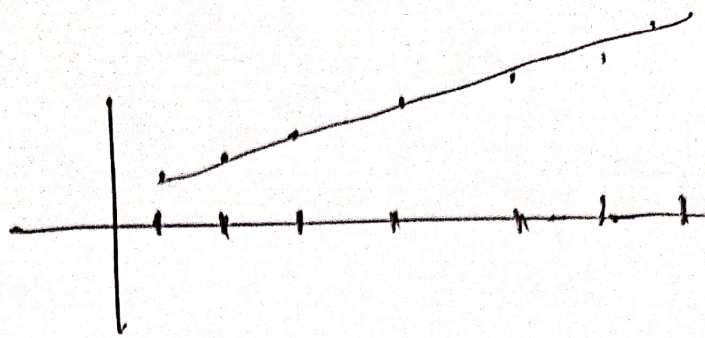
$$\therefore f(x) = 0 + 0.9(x - 0) - 0.335(x - 0)\left(x - \frac{\pi}{4}\right)$$

$$0 \leq x \leq \pi/2$$

(Ans)

$$3$$

$x(D)$	$\Delta n(y)$
1	0.5
2	2.5
3	2
4	4
5	4
6	3.5
7	6.5



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Yes, we can do a linear regression. ✓

$$\sum x = 28$$

$$\sum y = 24$$

$$\sum xy = 119.5$$

$$\sum x^2 = 140$$

$$\therefore y = a_0 + a_1 x$$

$$\text{So, } a_0 = 0.071$$

$$a_1 = 0.839$$

$$\therefore y = 0.071 + 0.839x$$

$$\therefore y(10) = 8.461$$

1805030
(04)

4

$$\begin{array}{l} x_0 = 1 \\ x_1 = 2 \\ x_2 = 3 \end{array} \quad \begin{array}{l} x_0 = 2 \\ x_1 = 3 \\ x_2 = 5 \\ x_3 = 7 \end{array} \quad \begin{array}{l} f(x_0) = 12 \\ f(x_1) = 19 \\ f(x_2) = 33 \\ f(x_3) = 51 \end{array}$$

$$\begin{aligned} \therefore f(x) = & \frac{x-3}{2-3} \times \frac{x-5}{2-5} \times \frac{x-7}{2-7} \times 12 \\ & + \frac{x-2}{3-2} \times \frac{x-5}{3-5} \times \frac{x-7}{3-7} \times 19 \\ & + \frac{x-2}{5-2} \times \frac{x-3}{5-3} \times \frac{x-7}{5-7} \times 33 \\ & + \frac{x-2}{7-2} \times \frac{x-3}{7-3} \times \frac{x-5}{7-5} \times 51 \end{aligned}$$

$$\therefore f(4) = 25.8 \quad \checkmark$$

1805030
(05)

5. Both the methods uses the concept of pivoting and elimination.

~~Q~~ In Gaussian method, a upper triangle is formed. Then solves using back substitute ✓

In Jordan method, the matrix is converted into an identity matrix. In this method, each index except the diagonal one needed to be converted to zero. Thus it requires more operations. ✓

1805030
(06)

b

$$v = 6.5$$

$$\therefore \sqrt{2gh} \tanh\left(\frac{\sqrt{2gh}}{2L} t\right) = v$$

$$\Rightarrow 4.42\sqrt{h} \tanh \frac{4.42\sqrt{h}}{9} t = 6.5 = f(t)$$

$$\text{let, } a = 0, b = 3.$$

init:

$$\text{mid} = 3/2 = 1.5.$$

$$f(\text{mid}), f(a)$$



1805030
(07)

7. $y = ax e^{bx}$

$\Rightarrow \frac{y}{x} = a e^{bx}$

$\Rightarrow \ln\left(\frac{y}{x}\right) = \ln a + bx$

let, $\frac{y}{x} = Y$

$\ln a = a_0$

$b = a_1$

$x = X$

So, the eqn is : $y = a_0 + a_1 x$

Now, $S_n = \sum (y - a_0 - a_1 x)^2$

$\therefore \frac{\partial S_n}{\partial a_0} = 2 \sum (y - a_0 - a_1 x) (-1)$

$\therefore \frac{\partial S_n}{\partial a_1} = 2 \sum (y - a_0 - a_1 x) \cdot (-x)$

$\therefore a_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$

$\therefore a_0 = \bar{y} - a_1 \bar{x}$

(from here)

$a = e^{a_0}$
and, $b = a_1$